

CONSOLIDATED ENGINEERING AND SOURCING CONSULTANTS (CESC)

Engineering White Paper

Transformer Online Condition Monitoring

DGA, Bushing Monitoring, OLTC Monitoring, Thermal Modelling, and Asset Health Interpretation

Prepared, reviewed, and issued by CESC

Website publication edition

Executive abstract

This paper summarizes how utilities can engineer transformer monitoring beyond isolated sensors by combining dissolved gas analysis, bushings, OLTC, loading, thermal models, and maintenance workflows into a practical asset-health program.

Document basis

Document type	Technical white paper
Primary basis	IEC 60599, IEC 60076 series, IEC 61850, IEEE C57.104...
Intended audience	Utilities, EPC contractors, industrial asset owners, and engineering teams
Publication note	Prepared in formal issuer style to demonstrate CESC engineering capability in studies, protection, digital integration, retrofit, and asset management.

Applicable references

Reference	Application in this paper
IEC 60599	Interpretation guidance for dissolved gases in transformer oil.
IEC 60076 series	Transformer design, operation, and thermal context.
IEC 61850	Reference framework relevant to the engineering subject of this paper.
IEEE C57.104	Complementary transformer dissolved-gas interpretation framework.
IEC 61936-1	Power installation engineering and general station design context.

Why condition monitoring matters

Large power transformers are long-life assets whose failure consequences extend beyond equipment replacement cost. Extended outage, fire risk, constrained system operation, and loss of redundancy can all outweigh the value of the transformer itself.

An effective monitoring strategy therefore links electrical diagnostics to maintenance decision-making rather than collecting data in isolation.

Monitoring philosophy

No single indicator is sufficient. DGA can identify developing thermal and electrical faults, but it should be interpreted together with loading trends, top-oil and winding temperature, moisture risk, OLTC operation count and signature, and bushing condition.

The preferred engineering model combines continuous measurements, event thresholds, rate-of-change logic, and asset-specific interpretation rules.

Engineering framework

A basic transformer health index can be structured around weighted condition groups: gases and gas rates, temperature and overload exposure, bushings, OLTC switching condition, oil quality, and known design history.

Where thermal modelling is used, the engineer should verify that operating duty remains within allowable hot-spot limits. A simplified relationship is often expressed as winding hot-spot temperature = ambient + top-oil rise + conductor hot-spot rise.

Engineering calculations and practical checks

Condition-monitoring systems are more credible when the monitoring philosophy is supported by simple engineering checks that operators and maintenance teams can verify. The examples below are intentionally practical and are presented in the same issuer-style format used throughout this CESC paper.

Example 1 - winding hot-spot temperature check

For a transformer operating at 42 C ambient temperature with a measured top-oil rise of 48 C and an estimated winding hot-spot gradient of 22 C, the simplified hot-spot temperature becomes: Hot-spot = $42 + 48 + 22 = 112$ C. This check is useful because alarm limits should not only reflect absolute top-oil temperature, but also the resulting winding thermal stress. Where the calculated hot-spot repeatedly approaches the transformer design limit, the asset manager should reassess loading profile, cooling effectiveness, and overload policy.

Example 2 - DGA rate-of-change assessment

Assume dissolved hydrogen increases from 120 ppm to 210 ppm over 30 days. The average rate is therefore $(210 - 120) / 30 = 3$ ppm/day. Even if the absolute value is not yet at a severe level, the trend may justify enhanced sampling because the engineering concern is the change trajectory rather than the number in isolation. This is particularly relevant when hydrogen, acetylene, ethylene, or methane movement coincides with loading or switching events.

Example 3 - transformer loading check against rating

For a 40 MVA transformer carrying 31 MVA actual load, percentage loading = $(31 / 40) \times 100 = 77.5$ percent. The result appears acceptable in isolation, but when combined with high ambient temperature, elevated top-oil rise, or cooling-stage failure, the same loading point may produce an unfavourable thermal condition. This is why CESC recommends that dashboards display transformer MVA, cooling status, top-oil temperature, and DGA trend together rather than as separate screens.

Example 4 - OLTC duty review

If an OLTC is designed for a maintenance inspection interval of 50,000 operations and the monitored counter already shows 38,000 operations, the consumed interval is $(38,000 / 50,000) \times 100 = 76$ percent. A fleet engineer should not wait for the absolute maintenance limit alone; operation rate, motor current signature, tap transition time, and diverter compartment condition should all be reviewed together. This converts simple counting into a true condition-based maintenance decision.

Engineering interpretation of these calculations

The practical value of these examples is not the arithmetic itself, but the discipline of linking sensor outputs to engineering action. Temperature checks support overload approval and cooling verification. Gas rate calculations support sampling escalation and fault investigation. Loading calculations support asset-utilization decisions. OLTC duty calculations support maintenance planning and outage coordination.

Practical diagnostics

DGA must be interpreted using both absolute gas values and trend rate. An apparently modest gas value may still represent a serious condition if the rate of increase is abnormal.

OLTC condition should not be left outside the monitoring scope. In many fleets, OLTC distress is one of the highest contributors to transformer forced outage and maintenance burden.

Bushing monitoring becomes especially valuable for high-voltage transformers where dielectric deterioration may not produce obvious operational symptoms before failure.

Execution and asset management value

The monitoring system should feed a maintenance workflow with clear escalation stages: normal review, enhanced sampling, focused inspection, planned outage, or immediate intervention. Without this workflow discipline, monitoring systems become data repositories rather than risk-reduction tools.

CESC recommends defining trigger matrices during engineering so that operators, maintenance teams, and asset managers act on the same technical criteria.

CESC engineering contribution

CESC can deliver monitoring philosophy, sensor selection, alarm logic, dashboard structuring, integration to station control systems, and practical maintenance decision rules that convert raw condition data into actionable asset-management guidance.

Conclusion

These website papers are structured to demonstrate how CESC approaches engineering subjects at system level, not only at equipment catalogue level. The intended message to clients is that CESC can define studies, design criteria, scheme architecture, implementation strategy, testing approach, and brownfield execution method across protection, control, digital integration, retrofit, and asset management scopes. In condition-monitoring applications specifically, CESC adds value by translating raw DGA, thermal, bushing, and OLTC data into engineering thresholds, decision rules, maintenance workflows, and integration logic that utilities can actually execute in service.